

Schmidt decomposition

a) $1427 = \frac{1027 + 1027 + 1207}{\sqrt{3}}$

$$1427 = \frac{1007 - 1027 + 1107 - 1127 + 2127}{2\sqrt{2}}$$

Justification

$$1427 = \sum_{ij} A_{ij} |i\rangle_A \otimes |j\rangle_B$$

Trivial case

$$1427 = \frac{1017 + 1027 + 1207}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} \left[107 \underbrace{(117 + 127)}_{\substack{\uparrow \\ \text{It should be Normalized}}} + 1207 \right]$$

$$1427 = \frac{1}{\sqrt{3}} \left(107 \left(\sqrt{2} \left(\frac{117 + 127}{\sqrt{2}} \right) \right) + 1207 \right)$$

$$= \frac{\sqrt{2}}{\sqrt{3}} 107 \otimes \left(\frac{117 + 127}{\sqrt{2}} \right) + \frac{1}{\sqrt{3}} 127 \otimes 107$$

$$= \frac{\sqrt{2}}{\sqrt{3}} 107_A \otimes \left(\frac{117 + 127}{\sqrt{2}} \right)_B + \frac{1}{\sqrt{3}} 127_A \otimes 107_B$$

$$= \frac{\sqrt{2}}{\sqrt{3}} 107_A \otimes \left(\frac{117 + 127}{\sqrt{2}} \right)_B + \frac{1}{\sqrt{3}} 127_A \otimes 107_B$$

$$1427 = \frac{1007 - 1027 + 1107 - 1127 + 2127}{2\sqrt{2}}$$

$$= \left(\frac{1007 - 1027}{2\sqrt{2}} \right) + \left(\frac{1107 - 1127}{2\sqrt{2}} \right) + \frac{2127}{2\sqrt{2}}$$

$$= \frac{107(107 - 127)}{2\sqrt{2}} + \frac{127(107 - 127)}{2\sqrt{2}} + \frac{127}{\sqrt{2}}$$

Each parts must be normalised

$$\begin{aligned}
 |\psi\rangle &= \frac{1}{2\sqrt{2}} \left(|0\rangle \left(\frac{1}{\sqrt{2}} \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right) \right) + \frac{1}{2\sqrt{2}} \left(|1\rangle \left(\frac{1}{\sqrt{2}} \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right) \right) + \frac{|2\rangle}{\sqrt{2}} \right) \right) \\
 &= \frac{1}{2\sqrt{2}} \left(|0\rangle \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right) + \frac{1}{2\sqrt{2}} \left(|1\rangle \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right) + \frac{|2\rangle}{\sqrt{2}} \right) \right) \\
 &= \frac{1}{2} \left((|0\rangle + |1\rangle) \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right) + \frac{|2\rangle}{2} \right) \\
 &= \frac{1}{2} \left[\left((|0\rangle + |1\rangle)_A \otimes \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right)_B \right) + \frac{|2\rangle_A \otimes |1\rangle_B}{2} \right] \\
 &= \frac{1}{2} \left[\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)_A \otimes \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right)_B \right] + \frac{|2\rangle_A \otimes |1\rangle_B}{2} \\
 &= \frac{1}{2} \left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right)_B \right] + \frac{|2\rangle_A \otimes |1\rangle_B}{2} \\
 &= \frac{1}{2} \left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \left(\frac{|0\rangle - |2\rangle}{\sqrt{2}} \right)_B \right] + \frac{|2\rangle_A \otimes |1\rangle_B}{2}
 \end{aligned}$$

b) Two-qubit maximal Entangled Entropy?

$$|\psi_{\max}\rangle_{AB} = \frac{1}{\sqrt{3}} \sum_{k=0}^2 |k\rangle_A \otimes |k\rangle_B$$

$$= \frac{1}{\sqrt{3}} (|0\rangle + |1\rangle + |2\rangle)_{AB}$$

What is the single qubit reduced State in case?

Solution

$$147_{AB} = \frac{1}{3} (1007 + 1117 + 1227)_{AB}$$

Partial trace of second matrix (B)

$$S_A = \text{Tr}_B [S_{AB}]$$

$$= \text{Tr}_B [147_{AB} \psi_{AB}]$$

$$= \text{Tr}_B \left[\frac{1}{3} (1007 + 1117 + 1227) \frac{1}{3} (1001 + 1111 + 1221) \right]$$

Only the following terms contributed for trace

10071001, 11171111 and 12271221, All the others are zero

$$S_A = \text{Tr}_B \left[\frac{1}{3} [10071001 + 11171111 + 12271221] \right]$$

$$= \frac{1}{3} [\text{Tr}_B [10071001] + \text{Tr}_B [11171111] + \text{Tr}_B [12271221]]$$

$$= \frac{1}{3} \left[\underbrace{100107}_{\frac{1}{3}} 107101 + \underbrace{111117}_{\frac{1}{3}} 117111 + \underbrace{121127}_{\frac{1}{3}} 127121 \right]$$

$$= \frac{1}{3} [107101 + 117111 + 127121]$$

$\underbrace{\hspace{10em}}_{I_3}$

$$S_A = \frac{I_3}{3}$$

⊙ Which probability distribution has the maximal (Shannon Entropy) 3

$$H(p) = - \sum_{i=1}^N p_i \log_2(p_i)$$

When the probabilities become equal we get maximum Shannon Entropy.

$$p_i = \frac{1}{N} \text{ for all } i$$

$$H_{\max}(p) = - \sum_{i=1}^N \frac{1}{N} \log_2\left(\frac{1}{N}\right)$$

$$H_{\max}\left(\frac{1}{N}\right) = - \sum_{i=1}^N \frac{1}{N} \log_2\left(\frac{1}{N}\right) = - \underbrace{N}_{\frac{N}{1}} \cdot \frac{1}{N} (-\log_2 N)$$

$$\underline{\underline{H_{\max}\left(\frac{1}{N}\right) = \log_2 N}}$$

c) purification $|\psi\rangle_{SE}$

Solution

$$|\psi\rangle_{SE} = \sum_{k=1}^Y \sqrt{\lambda_k} |k\rangle_S \otimes |k\rangle_E$$

Apply partial trace

$$S = \text{tr}_E(|\psi\rangle_{SE} \langle\psi|_{SE})$$

$$= \text{tr}_E \left(\sum_{j,k=1}^Y \sqrt{\lambda_j \lambda_k} |j\rangle_S \langle k|_S \otimes |j\rangle_E \langle k|_E \right)$$

$$= \sum_{j,k=1}^Y \sqrt{\lambda_j \lambda_k} |j\rangle_S \langle k|_S \otimes \text{tr}(|j\rangle_E \langle k|_E)$$

$$= \sum_{j,k=1}^Y \sqrt{\lambda_j \lambda_k} |j\rangle_S \langle k|_S \otimes (\delta_{jk})_E$$

$$\star S = \sum_{k=1}^Y \lambda_k |k\rangle_S \langle k|_S \otimes 1$$

$$= \sum_{k=1}^Y \lambda_k |k\rangle_S \langle k|_S$$

2. Single-qubit Gates

a) Given

$$n = \left[\frac{1}{\sqrt{2}} \ 0 \ \frac{1}{\sqrt{2}} \right] \text{ For } R_n(\pi)$$

$$R_n(\phi) = \exp \left[-i \left(\frac{\phi}{2} \right) n \cdot \sigma \right]$$

Required

$$iH = R_y\left(\frac{\pi}{2}\right) R_z(\pi) = R_n(\pi)$$

Solution

$$-iH = R_y\left(\frac{\pi}{2}\right) R_z(\pi)$$

$$= e^{-i \frac{\pi}{4} \sigma_y} e^{-i \frac{\pi}{2} \sigma_z}$$

$$= \left[\cos\left(\frac{\pi}{4}\right) I - i \sin\left(\frac{\pi}{4}\right) \sigma_y \right] \left[\cos\left(\frac{\pi}{2}\right) I - i \sin\left(\frac{\pi}{2}\right) \sigma_z \right]$$

$$= \left[\frac{1}{\sqrt{2}} I - \frac{i}{\sqrt{2}} \sigma_y \right] \left[0 \cdot I - i \sigma_z \right]$$

$$= \frac{1}{\sqrt{2}} \left[I - i \sigma_y \right] \left[-i \sigma_z \right] = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right] \left[-i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right]$$

$$= \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right] \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -i & -i \\ -i & i \end{bmatrix}$$

$$= -\frac{i}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = -i \underbrace{\left[\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right]}_H$$

$$= -iH$$

$$=$$

~~The minus sign not affect the measurement because~~
~~it is given these~~

$$\underline{\underline{-iH = -iH}}$$

$$R_n(\pi) = R_{\frac{1}{\sqrt{2}}}(n, 0, n)(\pi)$$

$$= e^{-i\frac{\pi}{2}} \frac{1}{\sqrt{2}} (2, 0, n) (\delta x, \delta y, \delta z)$$

$$= e^{-i\frac{\pi}{2}} \cdot \frac{1}{\sqrt{2}} (\delta x + \delta z)$$

$$= e^{-i\frac{\pi}{2}} \frac{1}{\sqrt{2}} (\delta x + \delta z)$$

$$= \cos\left(\frac{\pi}{2}\right) I - i \sin\left(\frac{\pi}{2}\right) \frac{1}{\sqrt{2}} [\delta x + \delta z]$$

$$= 0 - \frac{i}{\sqrt{2}} [\delta x + \delta z]$$

$$= -i \underbrace{\left[\frac{1}{\sqrt{2}} [\delta x + \delta z] \right]}_H$$

$$= \underline{\underline{-iH}}$$

Therefore

$$iH = \underbrace{R_y\left(\frac{\pi}{2}\right) R_z(\pi)}_{-iH} = \underbrace{R_n(\pi)}_{-iH}$$

$$\underline{-iH} = -iH = -iH$$

~~Blindly use the integral side with the value of it not correct~~
~~not a statement~~

$$\underline{\underline{-iH = R_y\left(\frac{\pi}{2}\right) R_z(\pi) = R_n(\pi)}}$$

b) Show that $HXH = Z$, $HYH = -Y$, $HZH = -X$

Solution

$$\begin{aligned} b) \quad HXH &= H \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \frac{2}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= I \end{aligned}$$

$HXH = I$

$$\begin{aligned} \checkmark \quad HYH &= H \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H \left(\frac{1}{\sqrt{2}} \begin{pmatrix} -i & i \\ i & i \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} -i & i \\ i & i \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} -i+i & i+i \\ -i-i & i-i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 2i \\ -2i & 0 \end{pmatrix} = \frac{-2}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= -Y \end{aligned}$$

$HYH = -Y$

$$\begin{aligned} \checkmark \quad HZH &= H \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1-1 & 1+1 \\ 1-1 & 1-1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = \frac{2}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= X \end{aligned}$$

$HZH = X$

c) Show that $HTH = R_x(\frac{\pi}{4})$

Solution

$$\begin{aligned} HTH &= H e^{-i \frac{\pi}{8} \sigma_z} H = e^{-i \frac{\pi}{8} \sigma_x} = R_x(\frac{\pi}{4}) \\ &= \underline{\underline{R_x(\frac{\pi}{4})}} \end{aligned}$$

d) Show that $HTHT = R_z(0)$

Solution $HTHT = e^{-i \frac{\pi}{8} \sigma_x} e^{-i \frac{\pi}{8} \sigma_z}$

$$UHT = e^{-i\frac{\pi}{8}x} e^{-i\frac{\pi}{8}z}$$

$$= \left[\cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} X \right] \left[\cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} Z \right]$$

$$= \cos^2 \frac{\pi}{8} I - i \cos \frac{\pi}{8} I \sin \frac{\pi}{8} Z - i \sin \frac{\pi}{8} X \cos \frac{\pi}{8} I +$$

$$- (i^2 \sin \frac{\pi}{8} X \sin \frac{\pi}{8} Z)$$

$$= \cos^2 \frac{\pi}{8} I - i \cos \frac{\pi}{8} I \sin \frac{\pi}{8} Z - i \sin \frac{\pi}{8} X \cos \frac{\pi}{8} I + \sin \frac{\pi}{8} X \sin \frac{\pi}{8} Z$$

$$= \cos^2 \frac{\pi}{8} I - i \cos \frac{\pi}{8} I \left(\sin \frac{\pi}{8} (X+Z) \right) + \sin \frac{\pi}{8} X \sin \frac{\pi}{8} Z$$

$$= \cos^2 \frac{\pi}{8} I - i \cos \frac{\pi}{8} I \left[\sin \frac{\pi}{8} (X+Z) \right] + \sin^2 \frac{\pi}{8} (XZ)$$

$$= \cos^2 \frac{\pi}{8} I - i \cos \frac{\pi}{8} I \left[\sin \frac{\pi}{8} (X+Z) \right] - i \sin^2 \frac{\pi}{8} Y$$

$$= \cos^2 \frac{\pi}{8} I - i \cos \frac{\pi}{8} I \left[\sin \frac{\pi}{8} (X+Z) \right] - i \sin^2 \frac{\pi}{8} Y$$

$$= \cos^2 \frac{\pi}{8} I - i \sin \frac{\pi}{8} \left(\cos \frac{\pi}{8} I (X+Z) + \sin \frac{\pi}{8} Y \right)$$

$$= \cos^2 \frac{\pi}{8} I - i \sin \frac{\pi}{8} \left(\cos \frac{\pi}{8} (X+Z) + \sin \frac{\pi}{8} Y \right)$$

b) CZ Gate and degrading channel

a) Solution

$$H = g | \text{int} \text{ (int)}$$

$$t = \frac{\pi}{g}$$

$$h = 1$$

$$e^A = 1 + \frac{A^1}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

$$e^A - 1 = \sum_{n=1}^{\infty} \frac{(A)^n}{n!}$$

Required

Show that

$$U(t) = e^{-iHt/\hbar} = C \tau$$

$$U(t) = e^{-iHt/\hbar}$$

$$U\left(\frac{\pi}{g}\right) = e^{-i(g|H\rangle\langle H|)\frac{\pi}{g}}$$

$$= e^{-i\pi(|H\rangle\langle H|)} = e^{-i\pi|H\rangle\langle H|}$$

$$= 1 + \frac{(-i\pi(|H\rangle\langle H|))}{1!} + \frac{(-i\pi|H\rangle\langle H|)^2}{2!} + \frac{(-i\pi|H\rangle\langle H|)^3}{3!} + \dots$$

$$= 1 + \frac{(-i\pi)}{1!}|H\rangle\langle H| + \frac{(-i\pi)^2}{2!}|H\rangle\langle H| + \frac{(-i\pi)^3}{3!}|H\rangle\langle H| + \dots$$

$$= 1 + \left(\frac{(-i\pi)}{1!} + \frac{(-i\pi)^2}{2!} + \frac{(-i\pi)^3}{3!} + \dots \right) |H\rangle\langle H|$$
$$\sum_{n=1}^{\infty} \frac{(-i\pi)^n}{n!}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{(-i\pi)^n}{n!} |H\rangle\langle H| = 1 + (|H\rangle\langle H|) \sum_{n=1}^{\infty} \frac{(-i\pi)^n}{n!}$$

$$= 1 + (|H\rangle\langle H|) (e^{-i\pi} - 1)$$
$$(e^{-i\pi} - 1)$$

$$= 1 + (|H\rangle\langle H|) (0 - 1) = 1 - |H\rangle\langle H| \quad \square$$

$$W(\mathbb{I}_g) = 2 + (1+2+2+1)(-2-2) = 2 + (1+2+2+1)(-2) \\ = 2 - 2(1+2+2+1)$$

$$= \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

$$W(\mathbb{I}_g) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix} = (2$$

b) Solution

1st Apply Hadamard on both qubit (qubit 1 and qubit 2) on $|00\rangle$

$$|142\rangle = (H_1 \otimes H_2) |00\rangle = |11\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

2nd - Apply C-2 on the $|142\rangle = |11\rangle$

$$|142\rangle = C2|11\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$$

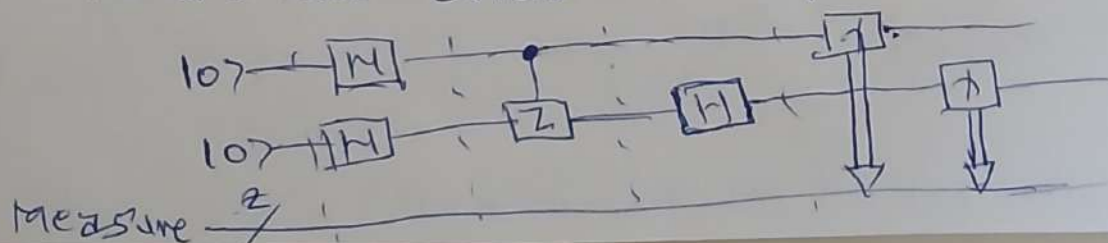
3rd Apply H on the second qubit $|142\rangle$

$$|143\rangle = (I_1 \otimes H_2) \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$$

$$= \frac{1}{2\sqrt{2}} (2|00\rangle + 2|11\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|143\rangle = \frac{(|00\rangle + |11\rangle)}{\sqrt{2}}$$

The Quantum circuit of $|143\rangle$ by using only C2 and H



c) Given

$$U = e^{-iHt}$$

$$S_S = S$$

$$S_R = 1 + 2H$$

$$H = g(1 + 2H^2)$$

$$A = g t$$

Restate

$$S' = ?$$

Solution

$$U = e^{-iHt}$$

$$= e^{-i g t (1 + 2H)}$$

$$= e^{-i g t} (1 + 2H)$$

$$= \left(e^{-i g t} - 1 \right) (1 + 2H) + I$$

$$= I + \left(e^{-i g t} - 1 \right) (1 + 2H)$$

$$S' = \text{tr} E (U (S \otimes (1 + 2H)) U^\dagger)$$

The required operators are

$$A_0 = \langle 0 |_2 U | 1 \rangle$$

$$= \langle 0 |_2 \left[I + \left(e^{-i g t} - 1 \right) (1 + 2H) \right] \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$= \frac{1}{\sqrt{2}} \left(\langle 0 |_2 I_2 | 0 \rangle + \langle 0 |_2 I_2 | 1 \rangle + \left(e^{-i g t} - 1 \right) \left[\langle 0 |_2 (1 + 2H) | 0 \rangle + \langle 0 |_2 (1 + 2H) | 1 \rangle \right] \right)$$

$$= \frac{1}{\sqrt{2}} \left[I_1 \underbrace{\langle 0 |_2 | 0 \rangle}_{=1} + I_1 \underbrace{\langle 0 |_2 | 1 \rangle}_{=0} + \left(e^{-i g t} - 1 \right) \left(\underbrace{1}_{=1} \langle 0 |_2 (1 + 2H) | 0 \rangle + \underbrace{0}_{=0} \langle 0 |_2 (1 + 2H) | 1 \rangle \right) \right]$$

$$= \frac{1}{\sqrt{2}} \left[I_1 + 0 + \left(e^{-i g t} - 1 \right) [0 + 0] \right] = \frac{1}{\sqrt{2}} I_1$$

HN

$$K_0 = \frac{1}{\sqrt{2}} I_1$$

$$K_1 = L_{12} U(1) + \dots$$

$$= \left[L_{12} \left(I_{12} + (e^{-i g t} - 1) |h\rangle\langle h| \right) \frac{1}{\sqrt{2}} (|0\rangle + |h\rangle) \right]$$

$$= \frac{1}{\sqrt{2}} \left[L_{12} I_{12} |0\rangle + L_{12} I_{12} |h\rangle + (e^{-i g t} - 1) L_{12} |h\rangle\langle h| |0\rangle + L_{12} |h\rangle\langle h| |h\rangle \right]$$

$$= \frac{1}{\sqrt{2}} \left[I_1 L_{12} I_2 |0\rangle + I_1 L_{12} I_2 |h\rangle + (e^{-i g t} - 1) (|h\rangle\langle h| L_{12} I_2 |0\rangle + |h\rangle\langle h| L_{12} I_2 |h\rangle) \right]$$

$$= \frac{1}{\sqrt{2}} \left[I_1 L_{12} I_2 |0\rangle + I_1 L_{12} I_2 |h\rangle + (e^{-i g t} - 1) (|h\rangle\langle h| L_{12} I_2 |0\rangle + |h\rangle\langle h| L_{12} I_2 |h\rangle) \right]$$

$$= \frac{1}{\sqrt{2}} \left[I_1 L_{12} |0\rangle + I_1 L_{12} |h\rangle + (e^{-i g t} - 1) (|h\rangle\langle h| L_{12} |0\rangle + |h\rangle\langle h| L_{12} |h\rangle) \right]$$

$$= \frac{1}{\sqrt{2}} \left[I_1 + (e^{-i g t} - 1) |h\rangle\langle h| \right]$$

$$= \frac{1}{\sqrt{2}} \left[|0\rangle\langle 0| + |h\rangle\langle h| + e^{-i g t} |h\rangle\langle h| - \underbrace{|h\rangle\langle h|}_{=X} \right]$$

$$= \frac{1}{\sqrt{2}} \left[|0\rangle\langle 0| + e^{-i g t} |h\rangle\langle h| \right]$$

$$\underline{\underline{||}}$$

Therefore

$$K_0 = \frac{1}{\sqrt{2}} I_1 = \frac{1}{\sqrt{2}} \left[|0\rangle\langle 0| + |h\rangle\langle h| \right]$$

$$K_1 = \frac{1}{\sqrt{2}} \left[|0\rangle\langle 0| + e^{-i g t} |h\rangle\langle h| \right]$$

Then!

$$S' = K_0 S K_0^+ + K_1 S K_1^+$$

$$= \frac{1}{\sqrt{2}} I_1 S \frac{1}{\sqrt{2}} I_1 + \frac{1}{\sqrt{2}} [I_0 \sigma_z I_0 + e^{i\theta t} I_1 \sigma_z I_1] S \frac{1}{\sqrt{2}} [I_0 \sigma_z I_0 + e^{i\theta t} I_1 \sigma_z I_1]$$

$$= \frac{1}{2} I_1 S I_1 + \frac{1}{\sqrt{2}} [I_0 \sigma_z I_0 + e^{i\theta t} I_1 \sigma_z I_1] S \frac{1}{\sqrt{2}} [I_0 \sigma_z I_0 + e^{i\theta t} I_1 \sigma_z I_1]$$

$$= \frac{1}{2} S + \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & e^{-i\theta t} \end{bmatrix} S \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta t} \end{bmatrix}$$

$$= \frac{1}{2} \left[\begin{bmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & e^{-i\theta t} \end{bmatrix} \begin{bmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta t} \end{bmatrix} \right]$$

$$= \frac{1}{2} \left[\begin{bmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{bmatrix} + \begin{bmatrix} S_{00} & S_{01} \\ S_{10} e^{-i\theta t} & S_{11} e^{-i\theta t} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta t} \end{bmatrix} \right]$$

$$= \frac{1}{2} \left[\begin{bmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{bmatrix} + \begin{bmatrix} S_{00} & S_{01} e^{i\theta t} \\ S_{10} e^{-i\theta t} & S_{11} (e^{-i\theta t} + e^{i\theta t}) \end{bmatrix} \right]$$

$$= \frac{1}{2} \left[\begin{bmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{bmatrix} + \begin{bmatrix} S_{00} & S_{01} e^{i\theta t} \\ S_{10} e^{-i\theta t} & S_{11} \end{bmatrix} \right]$$

$$= \frac{1}{2} \begin{bmatrix} 2S_{00} & S_{01} + S_{01} e^{i\theta t} \\ S_{10} e^{-i\theta t} + S_{10} & 2S_{11} \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 2S_{00} & S_{01} [1 + e^{i\theta t}] \\ S_{10} [1 + e^{-i\theta t}] & 2S_{11} \end{bmatrix}$$

$$S' = \begin{bmatrix} S_{00} & \frac{1}{2} S_{01} (1 + e^{i\theta t}) \\ \frac{1}{2} S_{10} (1 + e^{-i\theta t}) & S_{rr} \end{bmatrix}$$

Where,

Trig Identity

$$\boxed{\frac{(1 + e^{i\theta})}{2} = \cos\left(\frac{\theta}{2}\right) e^{i\frac{\theta}{2}}}$$

Then,

$$S' = \begin{bmatrix} S_{00} & S_{01} \cos(\theta t/2) e^{i\theta t/2} \\ S_{10} \cos(\theta t/2) e^{-i\theta t/2} & S_{rr} \end{bmatrix}$$

$$S' = \begin{pmatrix} S_{00} & S_{01} \cos(\theta t/2) e^{i\theta t/2} \\ S_{10} \cos(\theta t/2) e^{-i\theta t/2} & S_{rr} \end{pmatrix}$$

d) :

$$Z(s) = (2 + \gamma)S + PZSZ$$

$$N(s) = \begin{pmatrix} S_{00} & \cos(\theta t/2) e^{-i\theta t/2} S_{01} \\ \cos(\theta t/2) e^{i\theta t/2} S_{10} & S_{rr} \end{pmatrix}$$

Prove Estimation

$$M(P) = R_z(\theta t) Z(s) R_z^T(\theta t/2)$$

$$\phi = \frac{\pi - \cos(\frac{\theta t}{2})}{2}$$

Coherence factor of (FRHS)

$$FRHS = \cos(\frac{\theta t}{2}) e^{-\frac{\theta \theta t}{2}}$$

Target channel 1 target (RHS)

• Damping $\cos(\frac{\theta t}{2})$

• Rotation phase: $e^{-\frac{\theta \theta t}{2}}$

$$\pi - 2\phi = \pi - 2 \left(\frac{\pi - \cos(\frac{\theta t}{2})}{2} \right)$$

$$= \pi - \frac{2\pi - 2\cos(\frac{\theta t}{2})}{2}$$

$$= \pi - \frac{2\pi}{2} + \frac{2\cos(\frac{\theta t}{2})}{2}$$

$$= \pi - \pi + \frac{2\cos(\frac{\theta t}{2})}{2} = \cos(\frac{\theta t}{2})$$

$$= \underline{\underline{\cos(\frac{\theta t}{2})}}$$

Total factor.

$$FRHS = (\text{Damping}) @ (\text{Phase}) = \cos(\frac{\theta t}{2}) e^{-\frac{\theta \theta t}{2}}$$

Exactly mathematically equivalent

$$F_{LHS} = F_{RHS}$$

$$\cos\left(\frac{gt}{2}\right) e^{-i\frac{gt}{2}} = \cos\left(\frac{gt}{2}\right) e^{-i\frac{gt}{2}}$$

This prove that physical interaction also depolarizing channel with probability

$$p = \frac{1 - \cos\left(\frac{gt}{2}\right)}{2} \text{ induced by rotation } R_z\left(\frac{gt}{2}\right)$$